10.4 How Fields and Operators Transform

This note collects the transformation laws used from the first weeks of the course, with the same conventions as §10.3: $U(1+\delta\omega) = I + \frac{i}{2}\delta\omega_{\mu\nu}M^{\mu\nu}$.

Source and Target Manifolds

- **Source manifold**: the spacetime on which the coordinates x^μ are defined, here Minkowski space $\mathbb{R}^{1,3}$. Fields are functions on it.
- **Target manifold**: the space of values of a field. For a real scalar field it is $\mathbb{R}$; for a complex scalar field it is $\mathbb{C} \cong \mathbb{R}^2$. If the modulus $|\phi|$ is frozen (for example at low energies in a phase where the $U(1)$ symmetry is spontaneously broken), only the phase remains and the target is the circle S^1.
- **Scalar field**: a map ϕ : source manifold $\to$ target manifold whose value at a spacetime point does not depend on the observer.

Active and Passive Transformations

- **Active**: the field configuration is moved (rotated, boosted); the coordinate system is kept fixed.
- **Passive**: the observer's coordinates are relabeled, $x \to \bar{x} = \Lambda x$; the physical field is untouched.

Passive. Let Alice use coordinates x^μ and Bob $\bar{x}^\mu = \Lambda^\mu{}_\nu x^\nu$. For a scalar, both assign the same value to the same point P:

$$\bar{\phi}(\bar{x}) = \phi(x) \quad \Longrightarrow \quad \bar{\phi}(\bar{x}) = \phi(\Lambda^{-1}\bar{x}), \quad \text{i.e.} \quad \bar{\phi}(x) = \phi(\Lambda^{-1}x).$$

Active. In QFT, Lorentz transformations act on the Hilbert space through unitary operators $U(\Lambda)$, and a scalar field operator obeys

$$U(\Lambda)^{-1}\,\phi(x)\,U(\Lambda) = \phi(\Lambda^{-1}x).$$

Taking expectation values, $\langle\psi|U^{-1}\phi(x)U|\psi\rangle = \langle\psi|\phi(\Lambda^{-1}x)|\psi\rangle$: in the transformed state $U(\Lambda)|\psi\rangle$, the field at x has the value it had at $\Lambda^{-1}x$ in $|\psi\rangle$. The configuration has been moved by Λ.

The two agree. Writing $\phi'(x) \equiv U(\Lambda)^{-1}\phi(x)U(\Lambda)$, the active and passive results are the same function,

$$\phi'(x) = \bar{\phi}(x) = \phi(\Lambda^{-1}x),$$

so the two views make identical predictions. Moving the system by Λ is equivalent to relabeling the coordinates by Λ.

Example. A temperature field $T(\mathbf{x})$ on $\mathbb{R}^3$ has target $\mathbb{R}$. Rotating the system by R gives $T'(\mathbf{x}) = T(R^{-1}\mathbf{x})$: the temperature now at $\mathbf{x}$ is the one that used to be at $R^{-1}\mathbf{x}$.

Four-Vector Operators

Operators that carry a Lorentz index but no spacetime argument transform with Λ itself. The basic examples are the energy–momentum operator and the Lorentz generators:

$$U(\Lambda)^{-1} P^\mu U(\Lambda) = \Lambda^\mu{}_\nu P^\nu, \qquad U(\Lambda)^{-1} M^{\mu\nu} U(\Lambda) = \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma M^{\rho\sigma}.$$

Infinitesimally this is the commutator

$$[P^\mu, M^{\rho\sigma}] = i\left(g^{\mu\sigma} P^\rho - g^{\mu\rho} P^\sigma\right),$$

which, with $U(1+\delta\omega) = I + \frac{i}{2}\delta\omega_{\rho\sigma} M^{\rho\sigma}$, gives $U^{-1} P^\mu U = P^\mu + \delta\omega^\mu{}_\nu P^\nu$. Unlike $\phi(x)$, P^μ is not a function of spacetime: its *components* mix under Λ; there is no argument to transform.

Why Λ^{-1} for the Argument but Λ for Components?

- **Arguments**: the field at x after the transformation is the field that used to be at the point that was carried to x, namely $\Lambda^{-1}x$.
- **Components**: indices label directions, and directions are rotated or boosted along with the system, so they transform with Λ.
- **Both at once**: a vector field has an argument and a component index, $U(\Lambda)^{-1} V^\mu(x) U(\Lambda) = \Lambda^\mu{}_\nu V^\nu(\Lambda^{-1}x)$. The general case, with a spin part $S^{\mu\nu}$ for each representation, is in §10.3.

Object	$U(\Lambda)^{-1}(\cdot)\,U(\Lambda)$	**What transforms**
Scalar field $\phi(x)$	$\phi(\Lambda^{-1}x)$	the argument
Four-vector operator P^μ	$\Lambda^\mu{}_\nu P^\nu$	the components
Vector field $V^\mu(x)$	$\Lambda^\mu{}_\nu V^\nu(\Lambda^{-1}x)$	both

Transformation laws under $U(\Lambda)$

Consistency with the Group Composition Law

The rules must respect $U(\Lambda_1)U(\Lambda_2) = U(\Lambda_1\Lambda_2)$. Apply the two factors one after the other:

$$U(\Lambda_1\Lambda_2)^{-1}\phi(x)U(\Lambda_1\Lambda_2) = U(\Lambda_2)^{-1}\,\phi(\Lambda_1^{-1}x)\,U(\Lambda_2) = \phi(\Lambda_2^{-1}\Lambda_1^{-1}x) = \phi\big((\Lambda_1\Lambda_2)^{-1}x\big),$$
$$U(\Lambda_1\Lambda_2)^{-1}P^\mu U(\Lambda_1\Lambda_2) = U(\Lambda_2)^{-1}\,\Lambda_1{}^\mu{}_\nu P^\nu\,U(\Lambda_2) = \Lambda_1{}^\mu{}_\nu \Lambda_2{}^\nu{}_\rho P^\rho = (\Lambda_1\Lambda_2)^\mu{}_\rho P^\rho.$$

Both close correctly. Had we written $\phi(\Lambda x)$ for the scalar, the first line would give $\phi(\Lambda_2\Lambda_1 x)$, the wrong order. That is why the inverse appears in the argument.